Statistics
Parameter Estimation, Hypothesis Testing, Regression

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Lecture 5

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Parameter Estimation

- Many times, we have a model for the CDF of pdf of population feature of interest.
- The model CDF will have parameters.
- We will now discuss how to determine the **best estimate** of the paramter(s).
- As before we will obtain n independent sample realizations of the parameter of interest.

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What is being estimated?



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What is being estimated?



And what is the method?

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Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.
- Assume that the parameter set is θ.
- Question: What is the *likelihood* of observing of $x_1, \ldots x_K$ if the parameter value was θ?
  - Assume the population is Bernoulli with parameter p.
  - Assume the population is Poisson with parameter λ.
  - Assume the population is uniform in the interval $(0, a)$.
  - Assume the population is Gaussian with unit variance and mean μ.
  - Assume the population is exponential with parameter μ.

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Maximum Likelihood Estimate

- Find the θ that maximises the likelihood of the observations.

$$\hat{\theta} = \arg\max_{\theta} L(x_1, \ldots, x_K|\theta)$$

- Can also maximise the Log Likelihood function.
- Likelihood and log likelihood functions and MLEs
- Bernouilli

$$\begin{aligned} L(\underline{x}; p) &= \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^n(1-p)^{K-n} \\ \hat{p} &= \frac{\sum_{i=1}^{K} x_i}{K} \end{aligned}$$

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Maximum Likelihood Estimate

- Poisson

$$
\begin{aligned}
L(\underline{x};\lambda) &= \prod_{i=1}^{K} e^{-\lambda}\frac{\lambda x_i}{x_i!} = e^{-K\lambda}\frac{\lambda^{\sum_{i=1}^{K} x_i}}{\prod_{i=1}^{K} x_i} \\
\log L(\underline{x};\lambda) &= -K\lambda + \sum_{i=1}^{K} x_i \log\lambda - \log(\prod_{i=1}^{K} x_i) \\
\hat{\lambda} &= \frac{\sum_{i=1}^{K} x_i}{K}
\end{aligned}
$$

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Maximum Likelihood Estimate

- Gaussian with known σ and unknown μ.

$$
\begin{aligned}
L(\underline{x};\lambda) &= \prod_{i=1}^{K} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} = \frac{1}{(\sqrt{2\pi}\sigma)^K} e^{-\sum_{i=1}^{K}\frac{(x_i-\mu)^2}{2\sigma^2}} \\
\log L(\underline{x};\lambda) &= -K\log(\sqrt{2\pi}\sigma) - \sum_{i=1}^{K}\frac{(x_i-\mu)^2}{2\sigma^2} \\
\hat{\mu} &= \frac{\sum_{i=1}^{K} x_i}{K}
\end{aligned}
$$

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Maximum Likelihood Estimate

- MLE for Uniform in $(0, a)$

$$\begin{aligned} L(\underline{x}; \lambda) &= \frac{1}{a^K} \\ \hat{a} &= \max\{x_1, x_2, \ldots, x_K\} \end{aligned}$$

- Expectation of Estimate is the **bias** and variance is the mean squared error.

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More on estimation methods

- Best guess of a random variable before measurement
- What do I mean by best? Two possibilities: Most likely value and the one that makes least error. latter is further characterized as the one that minimizes mean squared error.
- 0-1 Bernoulli, Loaded die with $p_1 = 0.2$ and all other values having probability 0.16, Exponential with mean 1, Gaussian with mean 2 and variance 1.
- Typically, we have a belief about the parameter, expressed as a CDF (or PDF) for the parameter, say $F(\theta)$.
- After measurement, you have additional information. You can now change your belief based on the measurement using Baye's theorem.

$$F(\theta \mid x_1, x_2, \ldots, x_n) = \frac{F(x_1, x_2, \ldots, x_n \mid \theta) F(\theta)}{\int_\theta F(x_1, x_2, \ldots, x_n \mid \theta) F(\theta)\, d\theta}$$

- THis is called the Maximum a Posteriori (MAP) estimate

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MAP Estimate

Bernoulli with priors $\mathsf{Prob}(p = 0.3) = 0.5$ $\mathsf{Prob}(p = 0.5) = 0.2$ $\mathsf{Prob}(p = 0.7) = 0.3$. Determine the expression for the posterior based on five samples.

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